

The empirical recurrence for OEIS A187029, proved from an exact model

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Abstract

OEIS A187029 carries an empirical recurrence of order 3. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a finite sum over walk shapes of the ways each fits the board, from which a provably satisfies a MONIC linear recurrence of order $S = 10$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved: the objects counted are walks, but the count is a finite sum over their shapes and not a walk count in any digraph.

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1 The conjecture

OEIS A187029 is “Number of 4-step one or two collinear space at a time queen's tours on an $n \times n$ board summed over all starting positions.”. It has offset 1 and begins

0, 24, 1344, 7056, 19568, 39348, 66360, 100380, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 3504n^2 - 18540n + 24444$ for $n > 5$. $a(n) = 3a(n-1) - 3a(n-2) + a(n-3)$ for $n > 8$.

As of the “Last modified” line on the live entry (Apr 20 2018 06:08:47, revision 10) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a sum over a FIXED set of shapes

The number of steps is fixed at 4 cells, hence 3 moves, and it is the BOARD that grows. So the walks themselves do not depend on n at all. Let M be the 16 vectors the piece may move by, read from the entry's own words, and call two walks the same SHAPE when one is a translate of the other. There are exactly 3504 shapes: self-avoiding sequences of 4 distinct cells in $\mathbf{Z}^2$ with each consecutive difference in M , counted up to translation, which a finite search enumerates once and for all.

A shape occupies a bounding box with sides $(w_1, \dots, w_2)$, the range of its cells in each coordinate. Placing it on a board of side n means choosing where that box sits, and the choices in the coordinates are independent, so the shape occurs in exactly $\prod_i \max(0, n - w_i)$ positions and in no others. Distinct pairs (starting cell, walk) are exactly distinct pairs (shape, placement)

— a walk determines its shape and its translation, and conversely — so, with N_w the number of shapes of bounding box w ,

$$a(n) = \sum_w N_w \prod_{i=1}^2 \max(0, n - w_i).$$

This is exact for every n , not asymptotic, and it is not a walk count in any digraph: there is no transfer matrix here and no window.

3 The bound

Write w^* for the largest bounding-box side over all 3504 shapes. Once $n \geq w^*$ every factor above is nonnegative, so the truncation does nothing and $a(n) = \sum_w N_w \prod_i (n - w_i)$, a polynomial in n of degree 2. Hence $(z - 1)^3$ annihilates a from $n = 6$ onwards, and the annihilator in force on the whole sequence is $z^7(z - 1)^3$, monic of order $S = 10$. Both numbers are read off the enumerated shapes; neither is assumed. The model was then asked for terms beyond those the entry publishes and reproduced the published ones exactly.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j < S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^3 - \sum_i c_i z^{3-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n - i)$ is a linear combination of shifts of a , so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a , hence every shift of a , hence any linear combination of shifts; that is r . A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. $\square$

Theorem 2.

$$a(n) = 3a(n - 1) - 3a(n - 2) + a(n - 3)$$

for every $n > 8$.

Proof. The terms $a(0), \dots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 8 + 1$ onwards, and the run exceeds $S + 3$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 8$ the recurrence fails on the entry's own published terms, so no larger range holds. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 30 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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